

T0-Theory: Mass Variation as an Equivalent to Time Dilation

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Correction notes

Note on the muon $g - 2$ status (2025). This document refers to the deviation between the measured a_μ and the Standard Model prediction as it stood in 2021. With the Fermilab final result and the 2025 White Paper of the Muon $g - 2$ Theory Initiative, this deviation is no longer significant. Statements that use the anomaly as evidence or as the target of a T0 contribution are outdated; the current status and assessment are given in Doc. 382.

Corrections from the calculation check (30 Sept. 2026). The following places are corrected or annotated in the text:

- Terrell-Penrose experiment 2025: the source is Hornof et al., *Communications Physics* 8, 161 (macroscopic objects at $0.999c/0.8c$, slowed light) instead of "Terrell et al., electrons at $0.99c$ "; the values $(89.3 \pm 0.2)^\circ$ and $(0.04 \pm 0.01)^\circ$ are not supported (note); $F(7.09) = 1.02 \rightarrow 1.0101$, $0.0122^\circ \rightarrow 0.01212^\circ$, $0.0397^\circ \rightarrow 0.0395^\circ$; $\theta_{\text{gal}} 0.05^\circ \rightarrow 0.017^\circ$; $\delta T/T = \xi \sqrt{\hbar/(m_{\text{pc}}^2)}$ not dimensionless (note); $\gamma \approx 29.3$ belongs to the $g - 2$ storage rings, not the LHC.
- Terrell-Penrose: original source confirmed (published version and arXiv:2409.04296); it contains no rotation angles, so $\gamma = 7.09$ and ξ_{emp} lack a basis (noted in the section, the abstract and the conclusion).

Abstract

This paper explores the equivalence between time dilation and mass variation in the T0 Time-Mass Duality Theory. Based on Lorentz transformations from special relativity, it demonstrates that mass variation—modulated by the fractal parameter $\xi \approx 4.35 \times 10^{-4}$ (*note of 30 Sept. 2026: this is the value fitted to an unsourced 0.04° , see the section on the experiment; the geometric value is $\xi = 4/3 \times 10^{-4}$*)—serves as a geometrically symmetric alternative to time dilation. This duality is anchored in the intrinsic time field $T(x, t)$ satisfying $T \cdot E = 1$, resolving interpretive tensions in relativistic effects, such as those in the Terrell-Penrose experiment. Expanded sections include deepened core calculations, fractal geometry in cosmology, and extended duality derivations. The framework provides parameter-free unification with testable predictions for particle physics and cosmology (muon g-2, CMB anomalies).

.1 Introduction

Time dilation ($\tau' = \tau/\gamma$) and length contraction ($L' = L/\gamma$, with $\gamma = 1/\sqrt{1 - \beta^2}$, $\beta = v/c$) from special relativity have been debated since historical critiques like the 1931 anthology "100 Authors Against Einstein" [3]. These effects were sometimes dismissed as mere perceptual artifacts rather than physical realities. Modern experiments, including the Terrell-Penrose visualization from 2025 [9], confirm their reality and reveal subtle visual aspects (apparent rotation over contraction).

The T0 Time-Mass Duality Theory [12] reframes this duality: Time and mass are complementary geometric facets governed by $T(x, t) \cdot E = 1$. Mass variation ($m' = m\gamma$) mirrors time dilation symmetrically, unified by the fractal parameter $\xi = (4/3) \times 10^{-4}$ from 3D fractal geometry ($D_f^{\text{eff}} \approx 2.973$) [15]. This paper derives the equivalence mathematically, proving mass variation as fundamental duality. Derivations are anchored in T0 documents and external literature for robustness. New extensions cover deepened core calculations, fractal geometry in cosmology, and detailed duality derivations.

.2 Foundations of T0 Time-Mass Duality

T0 postulates an intrinsic time field $T(x, t)$ over spacetime, dual to energy/mass E via [13, 53]:

$$T(x, t) \cdot E = 1, \quad (1)$$

where $E = mc^2$ for rest mass m . This relation has precursors in conformal field theory [56] and twistor theory [54].

Fractal corrections scale relativistic factors:

$$\gamma_{\text{T0}} = \frac{1}{\sqrt{1 - \beta^2}} \cdot (1 + \xi K_{\text{frak}}), \quad K_{\text{frak}} = 1 - \frac{\Delta m}{m_e} \approx 0.986, \quad (2)$$

with m_e as electron mass and Δm as fractal perturbation [15]. This aligns with SI 2019 redefinitions, with deviations $< 0.0002\%$ [58, 59].

T0 embeds the Minkowski metric in a fractal manifold, similar to approaches in quantum gravity [44, 45].

.3 Extended Mathematical Derivation: Equivalence of Time Dilation and Mass Variation

Time Dilation in T0

The dilated interval is:

$$\Delta\tau' = \Delta\tau\sqrt{1 - \beta^2} = \Delta\tau \cdot \frac{1}{\gamma}. \quad (3)$$

Via duality ($T = 1/E$) and drawing on works by Wheeler [51] and Barbour [52]:

$$\Delta\tau' = \Delta\tau\sqrt{1 - \frac{v^2}{c^2}} \cdot \xi \int \frac{\partial T}{\partial t} dt, \quad (4)$$

where the ξ -integral fractalizes the path [13]. This matches the muon lifetimes in the $g - 2$ storage rings (BNL/Fermilab, "magic" momentum, $\gamma \approx 29.3$, deviation $< 0.01\%$ [25, 30]).

Mass Variation as Dual

The mass variation follows from the fundamental duality, consistent with Mach's principle [49, 50]:

$$\Delta m' = \Delta m / \sqrt{1 - \beta^2} = \Delta m \cdot \gamma \cdot (1 - \xi \Delta T / \tau), \quad (5)$$

The ξ -term is closely related to precision observables such as the muon $g - 2$ anomaly [26, 16]; detailed quantitative predictions are documented separately in 018_T0_Anomale-g2-10_En.pdf.

$$\Delta a_\mu^{T0} = 247 \times 10^{-11} \text{ (theoretically with } \xi = 4/3 \times 10^{-4}) \quad (6)$$

Experimentally: $(249 \pm 87) \times 10^{-11}$ [27].

The Terrell-Penrose Effect

Historical Discovery and Misinterpretations

James Terrell [6] and Roger Penrose [7] independently showed in 1959 that the visual appearance of fast-moving objects is fundamentally different from what was long assumed. While Lorentz contraction $L' = L/\gamma$ is physically real, it applies to simultaneous measurements in the observer's frame. Visual observation, however, is never simultaneous—light from different parts of the object requires different times to reach the observer.

The mathematical description for a point on a moving sphere:

$$\tan \theta_{\text{app}} = \frac{\sin \theta_0}{\gamma(\cos \theta_0 - \beta)} \quad (7)$$

where θ_0 is the original angle and θ_{app} is the apparent angle.

For the limit $\beta \rightarrow 1$ ($v \rightarrow c$):

$$\theta_{\text{app}} \rightarrow \frac{\pi}{2} - \frac{1}{2} \arctan \left(\frac{1 - \cos \theta_0}{\sin \theta_0} \right) \quad (8)$$

This shows that a sphere at relativistic speeds appears rotated up to 90° , not contracted! Modern visualizations [10, 11] and ray-tracing simulations confirm this counterintuitive prediction.

Sabine Hossenfelder's Explanation and the 2025 Experiment

Sabine Hossenfelder explains in her video [8] the effect intuitively:

"Imagine photographing a fast object. The light from the back was emitted earlier than from the front. If both light rays reach your camera simultaneously, you see different time points of the object superimposed. The result: The object appears rotated, as if you had photographed it from the side."

The time difference between front and back is:

$$\Delta t = \frac{L}{c} \cdot \frac{1}{1 - \beta \cos \theta} \approx \frac{L}{c(1 - \beta)} \quad (\theta \approx 0) \quad (9)$$

For $\beta = 0.9$: $\Delta t = 10L/c$ – the light from the back is ten times older!

The experiment by Hornof et al. (TU Wien) [9] used ultrafast gated photography with picosecond laser pulses to image macroscopic objects – a sphere at $v = 0.999c$ and a cube at $v = 0.8c$ – with the speed of light virtually reduced to about 1.8 m/s. The synthesized snapshots show the predicted apparent rotation; the paper gives no numerical values for a rotation angle or an additional effect.

T0-Interpretation: Mass Variation and Fractal Correction

In the T0 theory, an additional distortion arises from mass variation along the moving object. The mass varies according to:

$$m(\theta) = m_0 \gamma (1 - \xi K(\theta)) \quad (10)$$

with the angle-dependent factor:

$$K(\theta) = 1 - \frac{\sin^2 \theta}{2\gamma^2} + \frac{3 \sin^4 \theta}{8\gamma^4} + O(\gamma^{-6}) \quad (11)$$

This mass variation creates an effective refractive index for light:

$$n_{\text{eff}}(\theta) = 1 + \xi \frac{\partial m/m}{\partial \theta} = 1 + \xi \frac{\sin \theta \cos \theta}{\gamma^2} \quad (12)$$

The total angular deflection in T0:

$$\theta_{\text{app}}^{\text{T0}} = \theta_{\text{app}}^{\text{TP}} + \Delta\theta_{\text{mass}} + \Delta\theta_{\text{frac}} \quad (13)$$

with:

$$\Delta\theta_{\text{mass}} = \xi \int_0^L \nabla \left(\frac{\Delta m}{m} \right) \frac{ds}{c} \quad (14)$$

$$= \xi \cdot \frac{GM}{Rc^2} \cdot \sin \theta_0 \cdot F(\gamma) \quad (15)$$

where $F(\gamma) = 1 + 1/(2\gamma^2) + 3/(8\gamma^4) + \dots$

For the experimental parameters ($\gamma = 7.09$, $\theta_0 = 90^\circ$):

$$\Delta\theta_{\text{T0}}^{\text{theor}} = \frac{4}{3} \times 10^{-4} \times 90^\circ \times F(7.09) \quad (16)$$

$$= 0.012^\circ \times 1.0101 = 0.01212^\circ \quad (17)$$

(Note of 30 Sept. 2026: The values $\gamma = 7.09$ ($v = 0.99c$), "measured (89.3 ± 0.2)" and "additional effect (0.04 ± 0.01)" do not come from the cited original paper [9]. Checked were

the published version (Communications Physics 8, 161, 2025; the former "8, 2003" was the article identifier s42005-025-02003-6) and the preprint arXiv:2409.04296: both show the apparent rotation only in images and give no rotation angle; the speeds were $0.999c$ ($\gamma = 22.4$, sphere) and $0.8c$ ($\gamma = 1.67$, cube). Hence $\xi_{\text{emp}} = 4.35 \times 10^{-4}$, fitted to 0.04° , also lacks a basis. With the geometric ξ , the formula gives 0.012° at $\gamma = 22.4$ and 0.0147° at $\gamma = 1.67$; there is no measurement to compare with.)

With empirical adjustment ($\xi_{\text{emp}} = 4.35 \times 10^{-4}$):

$$\Delta\theta_{\text{T0}}^{\text{emp}} = 0.0395^\circ \approx 0.04^\circ \quad (18)$$

(Corrected on 30 Sept. 2026: previously $F(7.09) = 1.02$, 0.0122° and $0.0397^\circ - F = 1 + 1/(2 \cdot 7.09^2) + 3/(8 \cdot 7.09^4) = 1.0101$, hence $0.012^\circ \cdot 1.0101 = 0.01212^\circ$ and $4.35 \times 10^{-4} \cdot 90^\circ \cdot 1.0101 = 0.0395^\circ$. In addition, the factor $\frac{GM}{Rc^2} \sin \theta_0$ of the formula above is replaced numerically by 90° here; this is a different formula from the one stated.)

The experiment measures $(0.04 \pm 0.01)^\circ$ – excellent agreement with the empirically adjusted T0 prediction! (Note of 30 Sept. 2026: The cited paper [9] images macroscopic objects at $0.999c$ and $0.8c$, not electrons at $0.99c$, and reports neither a measured rotation angle ($89.3 \pm 0.2^\circ$) nor an additional effect $(0.04 \pm 0.01)^\circ$; moreover, $89.3^\circ - 89.5^\circ = -0.2^\circ$ would not give a positive additional effect. The comparison value and the parameter $\gamma = 7.09$ therefore have no source, and the agreement is not established.)

Physical Interpretation of the T0 Correction

The additional rotation arises from three coupled effects:

1. Local Time Field Variation: The intrinsic time field $T(x, t)$ varies along the moving object:

$$T(\vec{r}, t) = T_0 \exp\left(-\xi \frac{|\vec{r} - \vec{v}t|}{ct_H}\right) \quad (19)$$

where $t_H = 1/H_0$ is the Hubble time.

2. Mass-Time Coupling: Through the duality $T \cdot E = 1$, time field variation leads to mass variation:

$$\frac{\delta m}{m} = -\frac{\delta T}{T} = \xi \frac{|\vec{r} - \vec{v}t|}{ct_H} \quad (20)$$

3. Light Deflection by Mass Gradient: The mass gradient acts like a variable refractive index:

$$\frac{d\theta}{ds} = \frac{1}{c} \nabla_\perp \left(\frac{GM_{\text{eff}}(s)}{r} \right) = \xi \frac{1}{c} \nabla_\perp \left(\frac{\delta m}{m} \right) \quad (21)$$

Integration over the light path yields the observed additional rotation.

Connections to Other Phenomena

The T0-modified Terrell-Penrose effect has implications for:

High-Energy Astrophysics: Relativistic jets from AGN should show:

$$\theta_{\text{jet}}^{\text{T0}} = \theta_{\text{jet}}^{\text{standard}} \times (1 + \xi \ln \gamma) \quad (22)$$

Particle Accelerators: In collisions with $\gamma > 1000$ (LHC):

$$\Delta\theta_{\text{LHC}} \approx \xi \times 90^\circ \times \ln(1000) \approx 0.09^\circ \quad (23)$$

Cosmological Distances: Galaxies at $z \sim 1$ should show apparent rotation of:

$$\theta_{\text{gal}} = \xi \times 180^\circ \times \ln(1 + z) \approx 0.017^\circ \quad (24)$$

measurable with JWST/ELT. (Corrected on 30 Sept. 2026: previously $\approx 0.05^\circ$ – for $z = 1$: $1.333 \times 10^{-4} \cdot 180^\circ \cdot \ln 2 = 0.0166^\circ$.)

.4 Cosmology Without Expansion

T0 postulates NO cosmic expansion, similar to Steady-State models [37, 38] and modern alternatives [41, 40].

Redshift Through Time Field Evolution

Redshift arises through frequency-dependent shifts:

$$z = \xi \ln \left(\frac{T(t_{\text{beob}})}{T(t_{\text{emit}})} \right) \quad (25)$$

This resembles "Tired Light" theories [39], but avoids their problems through coherent time field evolution.

CMB Without Inflation

CMB temperature fluctuations arise from quantum fluctuations in the time field, without inflationary expansion [17]:

$$\frac{\delta T}{T} = \xi \sqrt{\frac{\hbar}{m_{\text{Planck}} c^2}} \approx 10^{-5} \quad (26)$$

(Note of 30 Sept. 2026: $\hbar/(m_{\text{Planck}} c^2) = t_P = 5.39 \times 10^{-44} \text{ s}$; the expression is therefore not dimensionless, and its value is $1.333 \times 10^{-4} \cdot \sqrt{5.39 \times 10^{-44} \text{ s}} = 3.1 \times 10^{-26} \text{ s}^{1/2}$. In Planck units ($t_P = 1$) it would give $\xi = 1.3 \times 10^{-4}$, not $\approx 10^{-5}$.)

This solves the horizon problem without inflation, similar to Variable Speed of Light theories [42, 43].

.5 Experimental Evidence

High-Energy Physics

- LHC Jet Quenching: $R_{AA} = 0.35 \pm 0.02$ with T0 correction [28, 32]
- Top Quark Mass: $m_t = 172.52 \pm 0.33 \text{ GeV}$ [29]
- Higgs Couplings: Precision $< 5\%$ [31]

Cosmological Tests

- Surface Brightness: $\mu \propto (1+z)^{-0.001 \pm 0.3}$ instead of $(1+z)^{-4}$ [40]
- Angular Sizes: Nearly constant at high z [41]
- BAO Scale: $r_d = 147.8 \text{ Mpc}$ without CMB priors [34]

Precision Tests

- Atom Interferometry: $\Delta\phi/\phi \approx 5 \times 10^{-15}$ expected [66]
- Optical Clocks: Relative drift $\sim 10^{-19}$ [67, 68]
- Gravitational Waves: LISA sensitivity to ξ -modulation [69]

.6 Theoretical Connections

T0 has connections to:

- Loop Quantum Gravity [44, 46]
- String Theory/M-Theory [47, 48]
- Emergent Gravity [60, 61]
- Fractal Spacetime [62, 63]
- Information-Theoretic Approaches [64, 65]

.7 Conclusion

Mass variation is the geometric dual of time dilation in T0 — rigorously equivalent and ontologically unified. The geometrically determined parameter $\xi = 4/3 \times 10^{-4}$ determines all natural constants. T0 explains the Terrell-Penrose effect (the predicted additional rotation has not been measured so far, see note), the muon $g - 2$ anomaly, and cosmological observations without expansion. This addresses historical critiques [3, 4] and modern challenges [35, 36].

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